Bragg's Law in the context of protein structure analysis

By Loredana-Maria Iacob

1. Introduction to the fundamental problem of determining atomic-scale crystal structures

Scientists lacked a clear understanding of how waves interact with the atomic planes in a crystal. Analysing diffraction peaks determines the spacing between atomic planes and, consequently, the crystal's structure. Numerous experiments and theories have been tested before a concise explanation of the interaction between X-Ray and atoms or plans of atoms have been found.

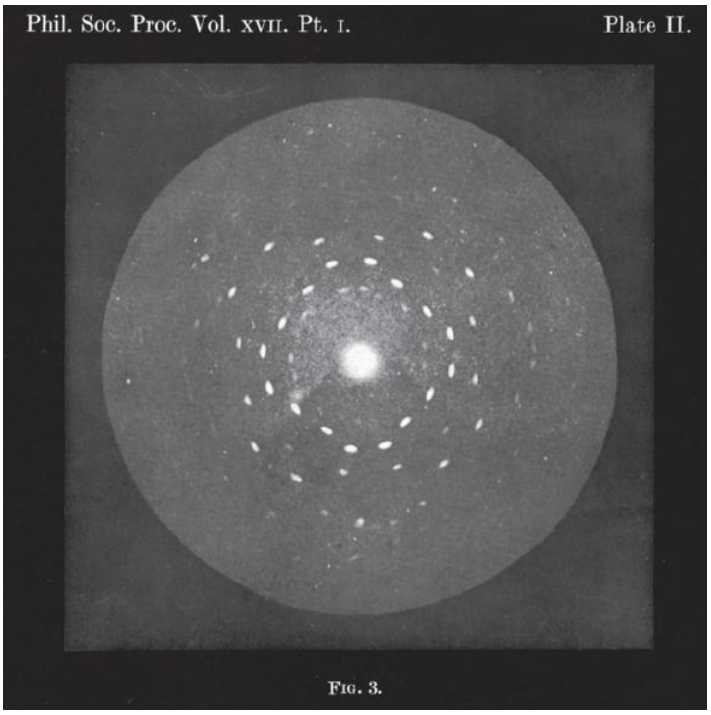
The father and son team of William Henry Bragg and William Lawrence Bragg were the first to attempt determining crystal structures using X-rays. They did this in 1913, utilizing X-ray diffraction to reveal how atoms are arranged within crystals. W. L. Bragg, inspired by Max von Laue and his experiments with X-Ray, proposed a corrected version of Laue's theory that he later used to prove Bragg's Law and form the basis of what we call today X-ray crystallography.

This discovery led to many of the most important scientific achievements of the last century, and these continue to the present day. This paper was the beginning of the field of X-ray crystallography, a subject that has enabled us to establish the complete structures of crystals, starting from the very simple to the most complex materials, such as proteins, viruses and the molecule that forms the very essence of life, namely DNA.

Around 20 or so Nobel Prizes have been awarded for research that has used the ideas described by Bragg. Modern genetics, medicine and the study of materials owe an incalculable debt to WLB, who at the astonishingly young age of 22, made a discovery that has changed our understanding of the world around us. Both father and son shared the 1915 Nobel Prize for their work, with WLB remaining to the present day the youngest Nobel Prize winner ever.

2. Bragg's experiments and observations

His paper starts with an overview of The Laue experiment, published in “Interference Phenomena with Räntgen Rays” published by Herren Friedrich *et. al.*, in which a beam of X-ray was passed through a crystal of simple structure and rectilinear pencils (narrow beams or rays of X-ray with a focused and controlled nature) were observed to spread in all directions forming a geometric pattern. The spots (where the maxima points hit the photographic plate) remain symmetric to the axes of a cubical zinc blade, when the ray is perpendicular to the third axis (Plate II, fig. 3). After further analysis of the plate and geometric relationships between the spots and crystal structures, it was noticed that the spots are points of intersection between ellipsis. Bragg explains the formation, shape and position of these ellipsis with a few geometric formulas.



Laue’s theory is that secondary X-rays are emitted by molecules, and these interfere to form a patter according to the system:

$$a\alpha = h_1\lambda$$

$$a\beta = h_2\lambda$$

$$a(1-\gamma) = h_3\lambda$$

Where h_1, h_2, h_3 are integers, a is the side of the cube, λ

the wavelength of the secondary waves and α, β, γ

, are the directional cosine of the new maxima. Note that the waves are parallel to the z axis so the surface of the beam parallel to the xy plane. Also, even though the system is technically correct, it was made with the wrong assumption that the atoms emit secondary waves, but these waves are in fact the ones reflected by the planes of the crystal, as it was later proved

This system suggests that all secondary waves are in phase in the direction of whose cosines are $\alpha,\ \beta,\ \gamma$

. The experimental set up allowed to calculate these values and prove the existence of such integers. However, not all h_1, h_2, h_3 are represented. Laue assumed that only certain λ

are represented because $\frac{\lambda}{a}$

also has to be an integer, but his assumption is wrong. Firstly, only certain λ

are present in the incident beam and, secondly, there are integers corresponding to certain $\frac{\lambda}{a}$

that do not appear in the pattern.

WLB makes the correct assumption that a continuous range of λ

are present and then proved, with the law he formulates, why such a beam does not form a uniform darkness. He suggests that X-rays are reflected by "centres of disturbance" (i.e. plans of atoms), the same way as light, and the reflected pulse interferes with the incoming ones to build a front.

Considering plans of atoms in the crystal, the reflected beams from each plane can be seen to interfere after being reflected to a θ

angle in planes found at distance d of each other. The wavelengths formed are λ

$$, \frac{\lambda}{2}$$

$$, \frac{\lambda}{3}$$

$$, \frac{\lambda}{4}$$

etc, where

$$\lambda = 2d\ cos\theta$$

.

The intensity of a spot depends on the "average breadth" of the beam, such us if $2d\ cos\theta$

is too small, pulses will neutralize each other and if it is too big, they will be too far away from each other. Thus, the reflections occur for specific combinations of $\lambda,\ d,\ \theta$

and this is why the plate does not show a uniform darkness.

Designing his experiment, WLB uses the Pope and Barlow theory to assume arrangements of atoms in a zinc blende and assumes than zinc and sulphur emit secondary waves of the same power to explain why some λ

miss in Laue's model. Taking the cubic packed model, with atoms at each corner and in the centre of each face, he considers $2a$ the distance between 2 atoms and defines 2 arbitrary points in the zx plane$A(pa, 0, qa)$

and in the yz plane $B(0, ra, sa)$

. A reflecting plane will pass through the origin and these 2 points, as shown in fig. 1. The directional cosines of such a plane are easy to calculate having defined p, q, r, s.

$$\dot{OC} = \dot{OA} \times \dot{OB} = (pa \quad 0 \quad qa) \times (0 \quad ra \quad sa) = (rq \quad ps \quad -pr)$$

$$l = \frac{rq}{\sqrt{p^2s^2 + q^2r^2 + p^2r^2}}$$

$$m = \frac{ps}{\sqrt{p^2s^2 + q^2r^2 + p^2r^2}}$$

$$n = \frac{-pr}{\sqrt{p^2s^2 + q^2r^2 + p^2r^2}}$$

Thus, the reflected beam will have the direction $2ln,\ 2mn,\ 2n^2 - 1$

, calculated using the formula of the reflected beam, where $\dot{R}$

is the reflected beam and $\dot{I}$

is incident (*that can be easily proven):

$$\dot{R} = \dot{I} - 2(\dot{N} \cdot I)\dot{I}$$

Notice how $n = \ cos\theta$

, where θ

is the angle of incidence. The corresponding wavelengths are known to be $\lambda = 2d$ $cos\ cos\ \ \theta$

, but it is easier to calculate the distance between 2 planes along the z axis l that the perpendicular distance d. Thus,

$$\lambda = 2d\ cos\ cos\ \ \theta = 2lcos\theta\ cos\theta = 2l\,n^2$$

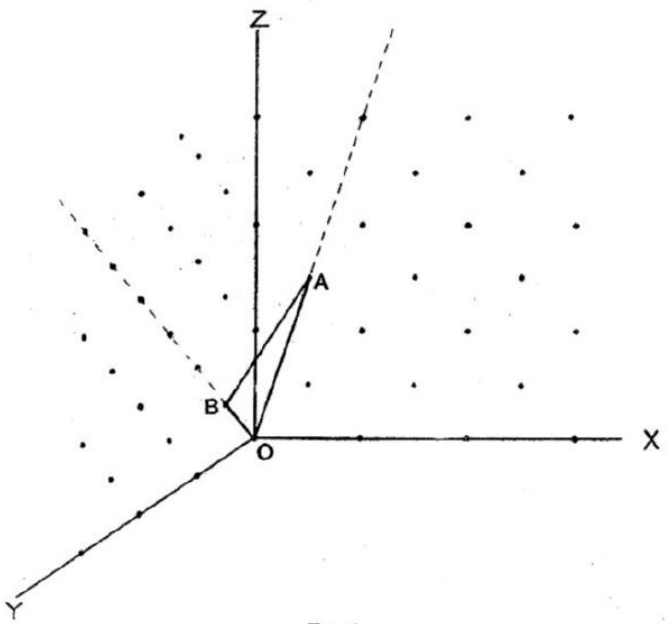


FIG. 1.

Another important point to be considered is the density of the atoms in the chosen plane. The strength of the reflected pulse depends on the number of atoms in a certain plane, so they need to be chosen accordingly (i.e. restriction on 2 of the parameters *p, r, l* defines the third). Consider the atoms as vertical columns parallel to the z axis. A plane with p=1 and r=1 will pass through one atom in each of these rows, therefore the next such plane will be at l=2a distance of this plane. For this set of planes $\lambda = 4an^2$

and the atom density is the greatest possible. For p=1 and r=2, such a plane passes through atoms in half of these vertical rows, so l also must be half as before to maintain the atom density constant. In general $l = \frac{2a}{l.c.m.\ of\ p\ and\ r}$

.

Furthermore, figure 3 and figure 4 show the pattern of spots and the planes that cause the interference. The planes that can be form by rotation around a certain line reflect light that form ellipsis on the photographic plate. By intersecting these ellipsis certain planes can be found, that contain 2 specific lines

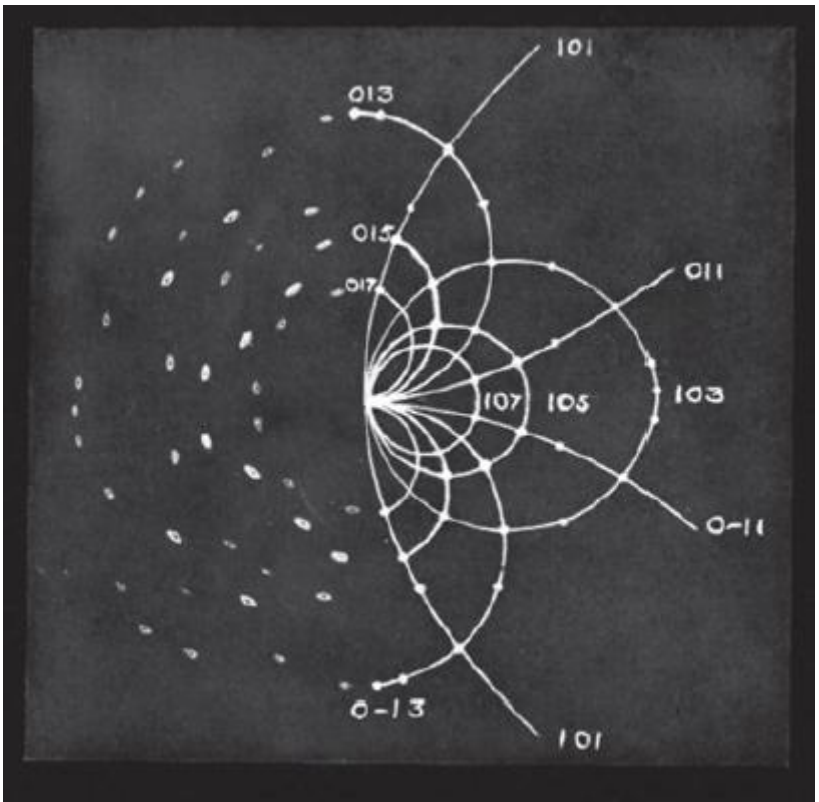


FIG. 4.

Notice how for a reflection to be observed, the coordinates must be all odd or all even. Having h_1, h_2, h_3 the phase at a certain atom, WLB describes

$$\frac{\lambda}{2a} = \frac{2h_3}{\sqrt{h_1^2 + h_2^2 + h_3^2}}$$

and his observations regarding this formula explain the intensity and the possible presence or absence of a certain spot, even though he admits that other explanations may also be possible. In his notes, he states that $\frac{a}{\lambda}$

must be in a certain range defined by the h values and within this range, the intensity of the spot is the highest in the middle and faints towards the edges. In order to prove this, he analysed a crystal in symmetric form and when it is tiled 3°. For a point in the plane defined by the points (0,0,0),

$$(3a, 0, a),$$

$$(0,3a, a)$$

the corresponding spot has a higher intensity in the tilled form $\left(\frac{\lambda}{a} = 6.5\right)$

than in the symmetric form$\left(\frac{\lambda}{a} = 4.3\right)$

.

It is also explained how no other arrangement of molecules offer a complete series of points and it also fits with other models for the ZnS blades.

The same analysis can be performed for other crystal arrangements to confirm these observations. He asses a simpler crystal to show the elasticity of the X-Rays and the intensity dependence on λ

, not only on the position of the plane.

In fig. 2 it is shown how the spots become more elliptical as the photographic plate is moved away from the crystal, proving that light is reflected and not channelled. This is probably the observation that made WLB realise the spots are formed by reflexion.

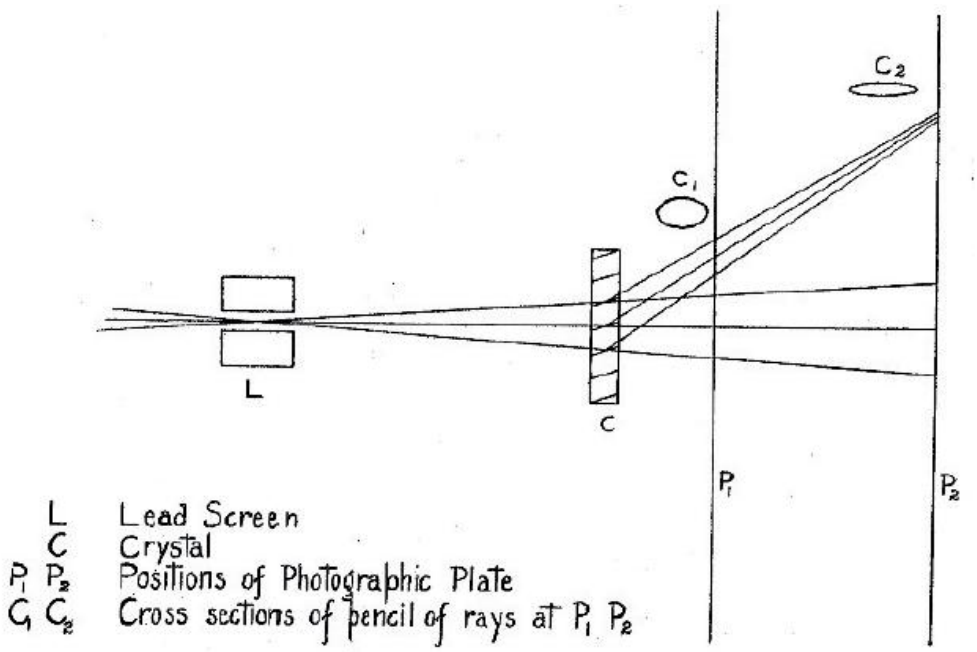


FIG. 2.

Further experiments with crystals show which parameters of the crystals are relevant for the formation and characteristics of the spots.

3. Bragg's Law statement and proof

Bragg's Law states "when the X-ray is incident onto a crystal surface, its angle of incidence, ϕ

, will reflect with the same angle of scattering, ϕ

. And, when the path difference, d is equal to a whole number, n, of wavelength, λ, constructive interference will occur.”. The wavelengths that are re-emitted after hitting a plane interact with each other in a constructive or destructive way, forming a diffraction pattern on photographic plate. The wavelengths that are reflected are proportionate to the distance between the planes and depend on the angle of incidence after the formula $n\lambda = 2d\ sin\phi$

, where d is the distance between planes, ϕ

the angle between the plane and the incident beam, and n an integer (note how in his paper he used the angle between the normal and the incident beam, so his original formula is $\lambda = 2d\ cos\theta$

).

Now, in order to prove this, multiple ways are possible. The geometric way refers to dependence of the phase on the wavelength of the beam and calculates the necessary angle of incidence. Another way uses Laue’s equation and the condition for elastic interactions to derive Bragg’s Law.

In a crystal, atoms are arranged in a certain patter, making its interaction with the X-ray predictable. Therefore, parts of the wave are spread in a different direction while others are passing in between. For the reflected waves to create a spot on the detector, they need to interfere constructively, which happens when they are in the same phase. This we know, from Laue's work, that happens when the path difference between different interacting atoms is an integer multiple of the wavelength. If the distance between 2 layers is d, the extra distance travelled is $2d\ sin\phi$

. This distance must be equal to $n\lambda$

. And thus, the Bragg's Law.

Taking a more algebraic look at this, we need to consider $\dot{k}$

the wave vector, with the norm $\frac{2\pi}{\lambda}$

, and λ

the wavelength of the particle. Laue's equation says:

$$\dot{k}' - \dot{k} = \dot{G}$$

Where $\dot{k}'$

is the reflected wavevector, $\dot{k}$

the incident vector, and $\dot{G}$

the reciprocal lattice vector. The difference vector needs to part of the crystal lattice for the interaction to be elastic (meaning that the 2 quotient spaces defined by the incident and reflected beam will be symmetric in rapport with the reflective plane). Connecting this to Bragg's Law and using vector algebra

$$\left|\dot{G}\right|^2 = \left|\dot{k'} - \dot{k}\right|^2 = \left|\dot{k}\right|^2 + \left|\dot{k'}\right|^2 - 2\left|\dot{k}\right|\left|\dot{k'}\right| \, cos \, cos \, 2 \, \phi$$

$$\left|\dot{G}\right|^2 = 2\left(\frac{2\pi}{\lambda}\right)^2 (1 - cos \, cos \, 2 \, \phi) = 4\left(\frac{2\pi}{\lambda}\right)^2 \phi$$

$$\left|\dot{G}\right| = \frac{4\pi}{\lambda} sin\phi$$

, but

$$\left|\dot{G}\right| = \frac{4\pi n}{d}$$

for a set of planes with distance d in between. Therefore

$$n\lambda = 2d \; sin\phi$$

Using Schrödinger's Equation offers a broader understanding of the Law and the way it is used for non-crystalline structures in nowadays medical research. It shows how a particle behaves after interacting with a periodic structure such as a crystal and makes the connection between the unknown structure and the known diffraction pattern.

Schrödinger's equation describes how waves behave when they encounter the crystal's atoms:

$$\left[\frac{-h^2}{2m}\nabla^2 + V(r)\right]\psi(r) = E\psi(r),$$

meaning that the energy of the wave is equal to the kinetic energy of the wave plus the potential energy of the atoms (its energy depends on its interactions (potential) and on how it moves (kinetic)).

Starting from the time independent form stated above, as the energy of the X-Ray photon does not depend on time in this context, and knowing that the potential is periodic due to the periodicity of the arrangement, we get the solution

$$\psi_{\dot{k}}(\dot{r}) = u_{\dot{k}}\psi(\dot{r})e^{i\dot{k}\cdot\dot{r}}$$

Where u_k

has the same periodicity as the crystal. This is the wave-like solution that allows diffraction to happen in the first place.

-TO BE REVISED

4. How to determine protein structure using X-Ray crystallography and Bragg's Law

Using X-Ray crystallography for determining structures gives mostly straightforward solutions with high precision, being a useful tool for further mechanistic understanding. Even though more complex than determining a crystal structure, proteins can also be solved with the same principles.

What we get from such an analysis is a diffraction pattern, as in fig. 5, and these patterns need to be interpreted. An electron density mas is made using Fourier transformation and finally, after many refinements made using plenty of statistical methods, the final protein structure is determined.

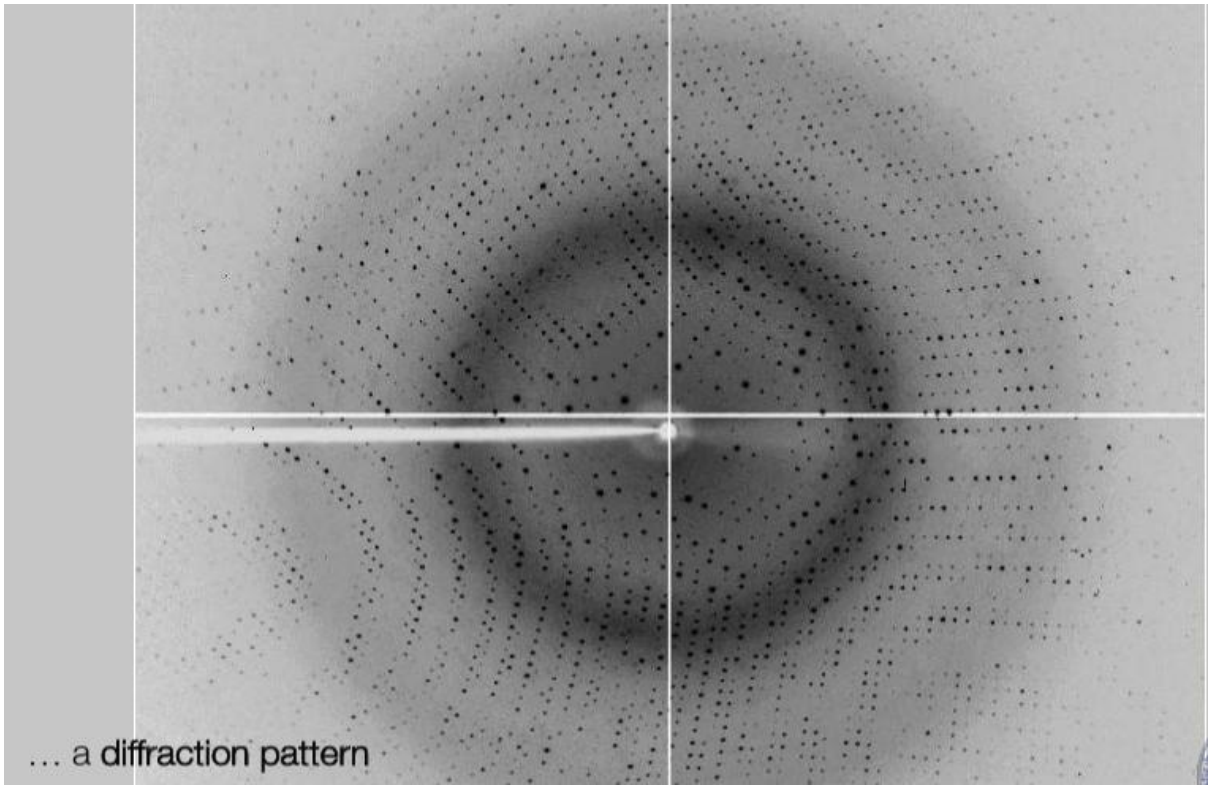


Fig. 5

The first step is to form a protein crystal, in order to make sure the structure is coherent and the x ray hits all the molecules from the same relative pozition. Next, Bragg's Law is used to determine the interplanar spacing between crystal planes. The distance is raported to the Miller indices hkl, which are specific to certain crystal arrangements and help solve the equation:

$$d_{hkl} = \frac{n\lambda}{2sin\theta}$$

The reflection angle θ

is calculated by a software that records the distance from the crystal to the detector *L* and the position *x where* the beam hits the detector

$$\theta = \frac{1}{2}\frac{x}{L}$$

Now, the electron density map can be modelled via Fourier transformation. The intensity of the diffraction spot is proportional to the sqaue of the amplitude of the scattered wave, an observation to which WLB also pointed. With *F(hkl)* being the structure factor, a complex number that represents the amplitude and the phase of the scattered X-Ray, f_j

the atomic factor (the probability of atom j to scatter X-Rays), and $(x_j,\ y_j,\ z_j)$

the coordinates of the atom j, the formula for the structure can be written like this:

$$F(hkl) = \sum_j \quad f_j e^{2\pi i(hx_j + ky_j + lzj)}$$

The electron density inside the unit cel is of course, a real-space function, and the diffraction pattern gives a Fourier transformation in its reciprocal space. This tells us how much the structure scatters the wave in each direction. The electron cloud is converted into a set of wave components that describes diffraction intensities. Performing an inverse Fourier transformation computes the real space picture of the structure:

$$F(hkl) = \iiint \quad \rho(x,y,z) \cdot e^{-2\pi i(hx+ky+lz)} dxdydz$$

becomes

$$\rho(x,y,z) = \frac{1}{V}\sum_{hkl} \quad F(hkl) \cdot e^{2\pi i(hx+ky+lz)+i\phi_{hkl}}$$

And for the last step, the actual reconstruction of the electron density map is done by inserting the measured values of the intensities in the inverse Fourier transformation. The only problem with this method is that the phase of the beam is unknown and different models have to be tested. Even though, theoretically, the phase problem is resolved with techniques such as molecular replacement (using the model of a protein with similar structure), multiple isomorphous replacement (crystals with and without heavy atoms are grown and the differences in diffraction pattern help solve the phase problem mathematically) or anomalous dispersion (some atoms scatter X-Ray differently depending on the wavelength, and data at different wavelengths form a system for solving the phases), the actual phase is more of a guessing game. For this reason, statistics also play a major role in determining protein structure.

Therefore, even in a world where mathematics is assumably absent such as biology and biochemistry, we can see that the most fundamental concept would have been unknown to humankind without developments in mathematics such as solutions to Schrödinger's equation, Fourier transformation and statistics.

Bragg's Law, which seems to play a minor role in this concept, is in fact the starting point of what we know today about atomic structures, and nothing would have been possible without WLB and his father's observations regarding crystallography.